INCH-POUND
MIL-PRF-3098/109F
7 January 2010
SUPERSEDING
MIL-PRF-3098/109E
19 December 2003

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR130/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; noncontrolled; antiresonance.

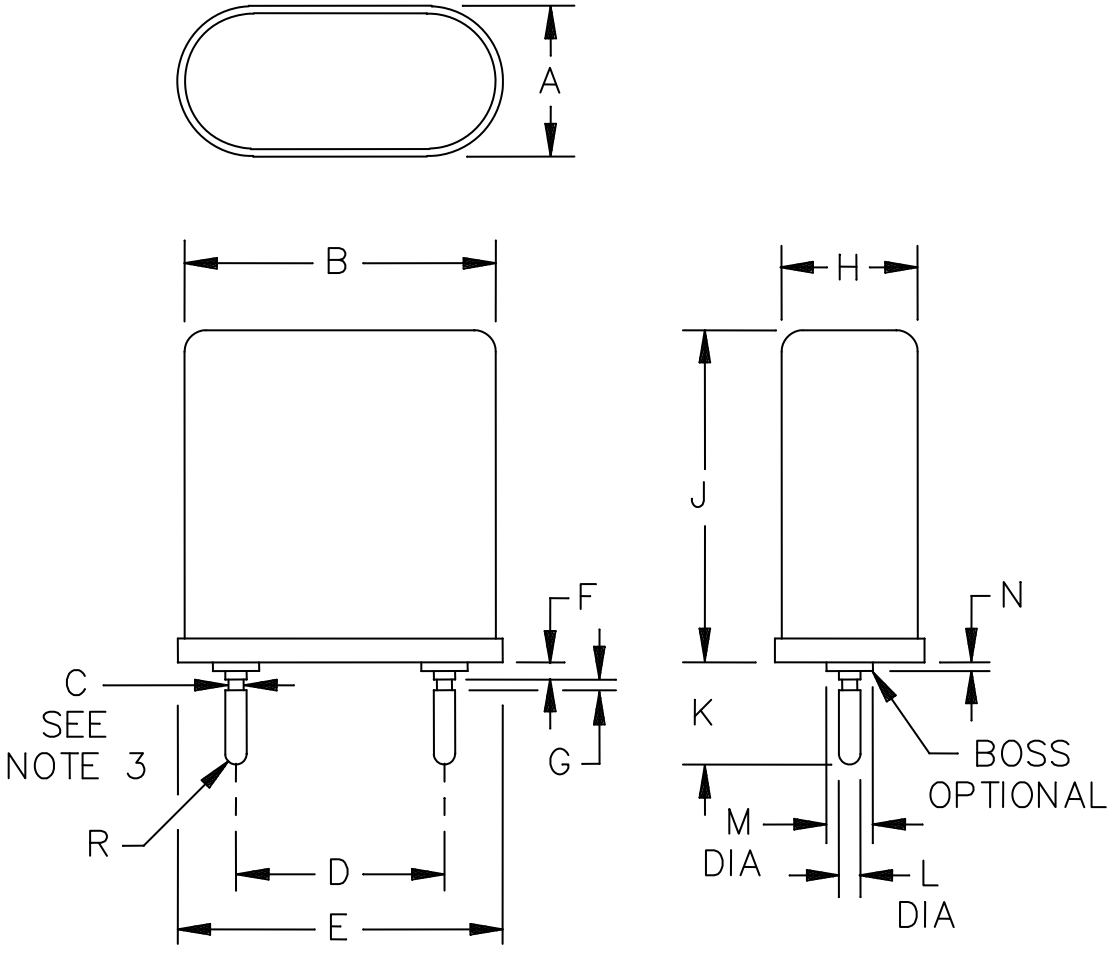

FIGURE 1. Crystal unit – CR130/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.352	---	8.94
B	---	.725	---	18.41
C	.030	.037	0.76	0.94
D	.460	.476	11.68	12.09
E	---	.757	---	19.23
F	.010	.070	0.25	1.78
G	.015	.025	0.38	0.63
H	---	.317	---	8.05
J	---	.775	---	19.68
K	.223	.248	5.66	6.30
L	.066	.216	1.68	5.49
M	.048	.052	1.22	1.32
N	.015	.025	0.38	0.63

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit – CR130/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20.0 MHz, inclusive.

Frequency tolerance (operating temperature range):

Primary: ±20 parts per million (ppm).

Secondary: ±30 ppm.

Equivalent resistance: See table I.

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TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.80 to 0.85	620
0.85+ to 0.90	600
0.90+ to 1.00	570
1.00+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.50	410
1.50+ to 1.62	370
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2.00	290
2.00+ to 2.12	270
2.12+ to 2.25	240
2.25+ to 2.60	190
2.60+ to 3.00	150
3.00+ to 3.40	110
3.40+ to 3.75	90
3.75+ to 4.00	75
4.00+ to 5.00	60
5.00+ to 7.00	35
7.00+ to 10.00	24
10.00+ to 15.00	22
15.00+ to 20.00	20

Mode of oscillation: Fundamental.

Temperature ranges:

Primary: -40°C to +90°C, inclusive.

Secondary: -55°C to -40°C and +90°C to +105°C, inclusive.

Rated drive level: 1.0 mW maximum.

Capacitance, shunt: 7.0 pF, maximum.

Antiresonance: load capacitance: 30.0 pF ±0.5 pF. 1/

1/ This crystal unit is similar to Crystal Type CR-85/U, except for operation at antiresonance and maximum equivalent resistance.

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Shock (specified pulse):	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Vibration: Method 204 of MIL-STD-202, test condition A.	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Thermal shock:	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Aging:

Frequency change permitted: ±5 ppm

Part or Identifying Number (PIN): CR130/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

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